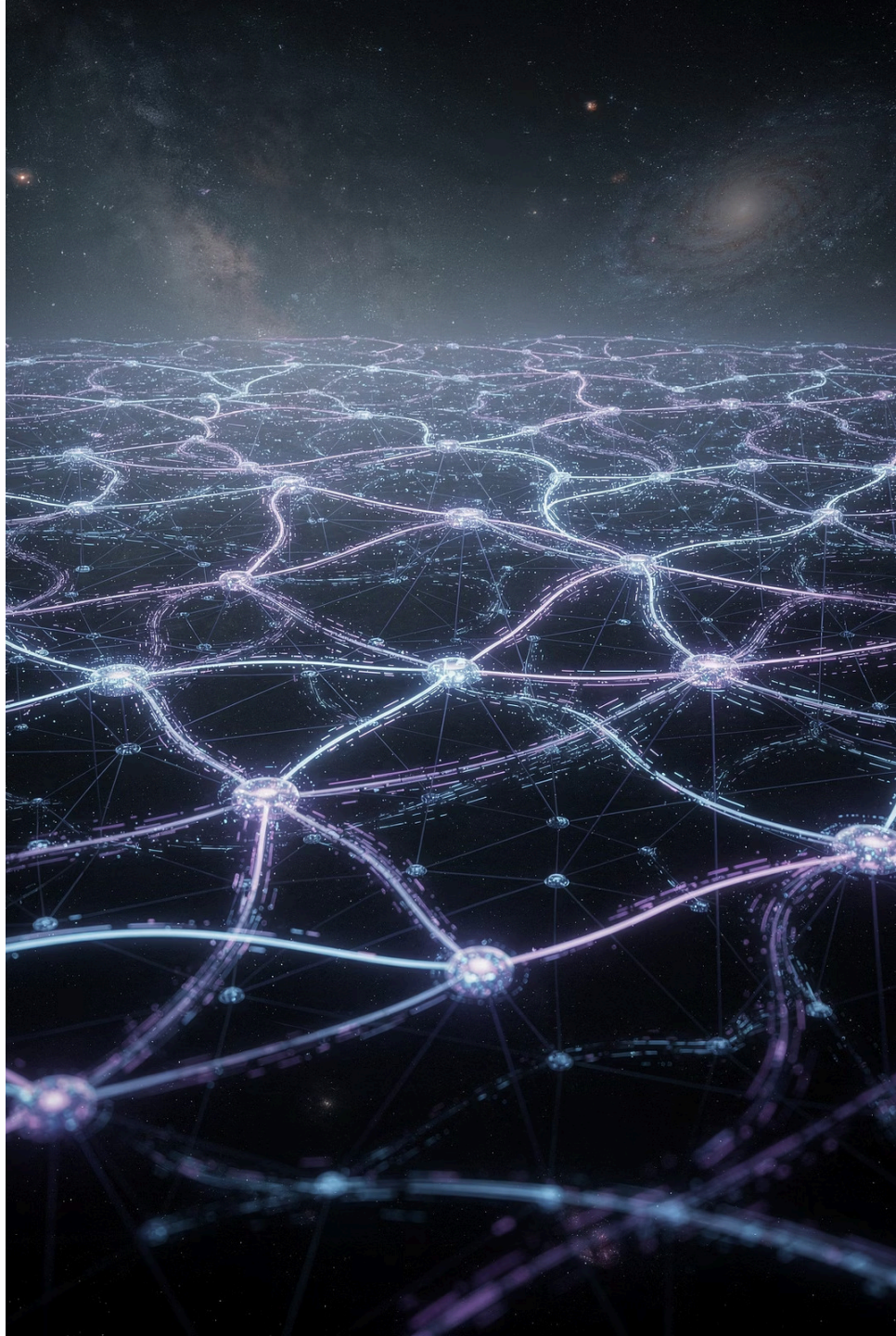
Foundations of AI & Generative AI

Building the Right Mental Models — Day 1

AI LITERACY SERIES



Day 1 Roadmap

What We'll Cover Today

Think of today as a guided tour through the AI landscape — from its origins to the cutting-edge models reshaping how we work. No prior AI experience needed. By the end, you'll have a clear mental model of how these systems actually work.

01

The AI Landscape

How AI evolved and where Generative AI fits in

02

Intro to Generative AI

What makes it different from traditional AI

03

Core Concepts

Tokens, embeddings, and probability — intuitively

04

Large Language Models

What makes LLMs "large" and how they learn

1.1 THE AI LANDSCAPE

How AI Grew Up: A 70-Year Journey

AI didn't appear overnight. It evolved through distinct eras — each building on the last, like floors of a skyscraper.

1

Rule-Based AI

1950s–80s. Humans wrote every rule. "If the patient has a fever AND cough, suggest flu."

2

Machine Learning

1990s–2010s. Systems learned patterns from data — like spam filters training on millions of emails.

3

Deep Learning

2010s. Neural networks with many layers unlocked image recognition and voice assistants.

4

Generative AI

2020s. Models don't just classify — they *create* text, images, code, and video.



Narrow AI vs. General AI vs. Generative AI

These three terms get confused all the time. Here's a simple way to think about them — using everyday analogies.

🎯 Narrow AI

One job, done brilliantly. Netflix's recommendation engine, Google Maps, spam filters, face unlock on your phone. Extremely capable — but only at its one task. Ask it to do something else and it fails completely.

🧠 General AI (AGI)

Human-level flexibility — still theoretical. AGI would reason, learn, and adapt across any domain, just like a person. We don't have it yet, and experts debate whether we're close or decades away.

✨ Generative AI

A creative narrow AI — flexible within language, images, and code. ChatGPT, Midjourney, and GitHub Copilot are all Generative AI. Impressive range, but still narrow: they generate — they don't truly think.

1.2 Introduction to Generative AI

What Is Generative AI, Really?

Traditional AI answers a question with a label: *"Is this email spam? Yes."* Generative AI goes further — it **produces new content** in response to a prompt.

Think of it as the difference between a **judge** and a **writer**. A judge evaluates; a writer creates. Generative AI is the writer.

Key characteristics that define Generative AI:

- It **produces original outputs** — text, images, audio, code, video
- It responds to **natural language prompts** — no coding required
- It generates **probabilistically** — the same prompt can yield different results
- It learns from **massive datasets** — books, websites, code repositories

Real-World Examples

- 📝 **Copilots** — Microsoft 365 Copilot drafts emails, summarizes meetings
- 💬 **Assistants** — ChatGPT, Claude answer questions conversationally
- 🎨 **Image gen** — Midjourney, DALL·E create visuals from text
- 💻 **Code gen** — GitHub Copilot writes and explains code
- 🎬 **Video** — Sora, Runway generate short video clips from prompts

Generative vs. Discriminative Models

This distinction is foundational. Both types of AI learn from data — but they answer very different questions.



Discriminative Models

Question they answer: "Which category does this belong to?"

Examples: Email spam detector, image classifier ("cat or dog?"), sentiment analysis ("positive or negative review?"). They draw boundaries between categories but cannot create anything new.



Generative Models

Question they answer: "What should come next — or what could be created?"

Examples: ChatGPT writing a report, Midjourney drawing a logo, GitHub Copilot completing your code. They learn the *distribution* of data, then sample from it to create new outputs.

A single product can combine both: a content moderation tool might use a discriminative model to flag harmful content AND a generative model to suggest safer rewrites.

1.3 CORE CONCEPTS

The AI Pipeline: From Raw Data to Smart Output

Every AI system — from a simple spam filter to GPT-4 — follows the same four-stage pipeline. Think of it like a factory assembly line.

📦 Data

Raw material. Text from billions of web pages, books, and code. Garbage in = garbage out — data quality determines model quality.

🏗️ Model

The blueprint. A mathematical structure (neural network) designed to find patterns. Architecture choices shape what the model can and can't do.

🎓 Training

The schooling. The model processes data, makes predictions, measures errors, and adjusts itself. Repeated billions of times. Very expensive — GPT-4 training cost ~$100M.

⚡ Inference

The job. Using the trained model to respond to new prompts. This is what happens every time you type a message to ChatGPT.

Language as Patterns — Not True Understanding

This is one of the most important mental models in all of AI. LLMs don't "understand" language the way you do. Here's what's actually happening — and why that matters.

How Humans Process Language

You read "The cat sat on the mat" and instantly picture a scene, feel the texture of the mat, understand cause and effect, and can reason about what happens next. You ground words in real-world experience.

How LLMs Process Language

An LLM sees billions of word sequences and learns: *"After 'The cat sat on the ___', the word 'mat' appears very frequently."* It learns statistical patterns — relationships between tokens — with no grounding in the physical world.

Tokens & Embeddings — Simply Put

- **Token:** The smallest unit the model reads. "Unbelievable" → ["Un", "believ", "able"]. GPT-4 reads ~100 trillion tokens during training.
- **Embedding:** A list of numbers that represents a word's meaning. "King" − "Man" + "Woman" ≈ "Queen" — the famous example of vector math on embeddings.
- **Why it matters:** The model knows that "Paris" and "France" are related not because it visited France, but because they co-occur in similar contexts millions of times.

⚠ This is why LLMs can "hallucinate" — they generate plausible-sounding patterns even when factually wrong.

Probability-Driven Generation

Every word an LLM writes is chosen through a probability calculation — not a lookup table, not a script. Understanding this unlocks intuition about why AI behaves the way it does.

1 You send a prompt

"Write me a subject line for a sales email about a new project management tool."

2 The model scores every word in its vocabulary

For the first word alone, it calculates a probability score for all ~50,000 tokens. "Boost" might score high. "Elephant" scores near zero.

3 It samples — not just picks the top word

A setting called **temperature** controls randomness. High temperature = creative and surprising. Low temperature = predictable and precise. That's why the same prompt gives different answers.

4 This repeats for every single token

One word at a time, left to right, until the response is complete. A 200-word answer involves 200+ separate probability decisions.

1.4 LARGE LANGUAGE MODELS

What Makes LLMs "Large" — and How They Learn

What Makes Them "Large"?

Three dimensions of scale — all of which grew dramatically after 2020:

- **Parameters:** The adjustable "knobs" in the network. GPT-2 had 1.5B. GPT-4 is estimated at ~1 trillion. More parameters = more capacity to learn nuance.
- **Training data:** Hundreds of billions of words — books, Wikipedia, GitHub, Reddit, and much of the public web.
- **Compute:** Thousands of specialized GPUs running for months. This is why only a handful of organizations can train frontier models.

Pretraining vs. Fine-Tuning

Pretraining is like going to school for 20 years — the model reads the internet and develops broad general knowledge. It learns to predict the next token across every topic imaginable.

Fine-tuning is like an internship — you take the trained model and expose it to a specialized dataset (e.g., medical records, legal documents, your company's support tickets) to make it expert in a narrow domain. Much cheaper than training from scratch.

Capabilities vs. Limitations

- ✅ Summarization, drafting, translation, code generation, Q&A
- ❌ Real-time data, true reasoning, guaranteed accuracy, private knowledge

Key Takeaways from Day 1

You've covered a lot of ground. Here are the five mental models to carry with you — the ones that will make every future AI conversation make more sense.

AI Evolved in Stages

Rule-based → ML → Deep Learning → Generative AI. Each era unlocked new capabilities by removing human bottlenecks.

Generative AI Creates

It doesn't classify — it produces. Text, code, images, and video — all from a simple natural language prompt.

Data → Model → Train → Infer

Every AI system follows this pipeline. Understanding it helps you ask better questions about any AI product.

It's Pattern + Probability

LLMs don't understand — they predict. This explains both their power (fluency) and their weakness (hallucination).

Pretrain Once, Fine-tune Many

The expensive general training happens once. Fine-tuning adapts that foundation to specific domains affordably.

Up next: Day 2 — Prompt Engineering, RAG, and Building AI-Powered Applications